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Effect of Solution pH on Protein Transmission and Membrane Capacity During Virus Filtration

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ABSTRACT: Although virus filtration is now an integral part of the overall viral clearance strategy for the purification of many commercial therapeutic proteins, there is currently little understanding of the factors controlling the performance of the virus filters. The objective of this study was to examine the effects of solution pH on protein transmission and capacity during virus filtration. Data were obtained using bovine serum albumin as a model protein with Viresolve 180 membranes oriented with both the skin-side up and skin-side down. Membranes were also characterized using dextran sieving measurements both before and after protein filtration. Membrane capacity and protein yield were minimal at the protein isoelectric point, which was due to the greater degree of concentration polarization associated with the smaller protein diffusion coefficient at this pH. In contrast, the actual protein sieving coefficient was maximum at the protein isoelectric point due to the absence of any strong electrostatic exclusion under these conditions. The yield and capacity were both significantly greater when the membrane was oriented with the skin-side down. These results provide important insights into the effects of solution conditions on the performance of virus filtration membranes for protein purification.

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KEYWORDS: virus filtration; protein purification; concentration polarization; fouling; membrane

Introduction

Insuring adequate viral clearance is an absolutely essential part of the downstream purification process for all protein-based therapeutics. Regulatory agencies require the use of a combination of inactivation/removal techniques, for example, low pH (Brorson et al., 2003), solvent–detergent treatments (Horowitz et al., 1985), and γ or UV irradiation (Roberts, 1994). Virus filtration provides a robust size-based technique for removal of viral contaminants, and this has now become a fairly standard component of most downstream purification processes (Burnouf and Radosevich, 2003; Shukla et al., 2007).

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An effective virus filter must provide high levels of viral clearance while insuring adequate protein transmission/recovery and good capacity. When originally developed, virus filtration was performed using tangential flow filtration with the membrane oriented so that the virus-retentive skin layer faced the feed. Current applications of virus filtration use disposable normal flow filters, with the more open side of the membrane facing the feed stream (Brough et al., 2002). This orientation provides greater capacity than obtained with the skin-side up, which is usually attributed to the capture of protein aggregates and other large foulants within the macroporous substructure thereby protecting the virus-retentive skin layer (Syedain et al., 2006).

The overall performance of virus filters is governed in large part by membrane fouling, which can alter the flux, capacity, and transmission characteristics. Higuchi et al. (2001, 2002) observed a large increase in flux and capacity for the Planova virus filters when operated at high NaCl concentrations or after treatment of the feed with DNase. The authors attributed these improvements to dissociation of large complexes formed between trace amounts of DNA and the γ-globulins. Similar results were reported during filtration of bovine serum albumin (BSA) solutions (Higuchi et al., 2004).

Syedain et al. (2006) examined the effect of operating conditions and membrane orientation on BSA fouling during virus filtration. The capacity for the Viresolve 180 membrane with the skin-side up increased with decreasing pressure and when operated at low flux (for constant flux experiments). The capacity was also much greater when the membrane was used with the skin-side down, which the authors attributed to the removal of protein aggregates within the macroporous substructure, thereby reducing the rate of fouling of the membrane skin. Similar results were obtained by Ireland et al. (2004) for normal flow parvovirus filters during filtration of a 10 g/L human IgG solution, with changes in the filter capacity attributed to variations in the properties and quantity of large protein aggregates at different solution pH and ionic strength.

In contrast to these results, Bolton et al. (2006) showed that prefiltration of a human IgG solution to remove protein aggregates had little effect on the capacity of parvovirus filters, suggesting that large aggregates were not a critical factor governing protein fouling during virus filtration. However, significant improvements in performance were obtained using a prefilter containing diatomaceous earth, which led the authors to hypothesize that fouling was due to a trace hydrophobic species that may have been a denatured form of the native protein.

In addition to these studies of protein fouling during virus filtration, there is also extensive prior work on protein fouling during both ultrafiltration and microfiltration. Fane et al. (1983) found that the filtrate flux during BSA filtration through both fully retentive and semi-permeable ultrafiltration membranes was minimum at the protein isoelectric point (p*I*), that is under conditions where the proteins had no net charge. The flux also decreased with increasing ionic strength for solutions above or below the protein isoelectric point, with this behavior attributed to the effects of solution environment on the extent of protein adsorption within the membrane pores and/or deposition on the membrane surface. Similar results were obtained by Palecek et al. (1993) for the constant pressure microfiltration of BSA, with the steady-state flux increasing with the square of the protein charge, clearly demonstrating the importance of electrostatic interactions.

Protein fouling of larger pore size microfiltration membranes is typically dominated by the deposition of large protein aggregates, with the filtrate flux determined by the properties of the protein deposit that forms on the membrane surface (Kelly et al., 1993). The deposited layers of BSA have been shown to function as compressible porous media, providing significantly greater resistance to filtration at high pressures (Opong and Zydney, 1991). Ho and Zydney (1999) found that the rate of protein fouling during filtration with isotropic membranes was much greater than that with asymmetric membranes due to the highly interconnected pore structure in the isotropic membranes. The data were in good agreement with the behavior predicted by a fouling model that accounted for the combined effects of pore-blockage and cake-filtration (Ho and Zydney, 2000).

Although these studies have provided useful insights into the fouling characteristics of different membranes, there is still considerable discrepancy regarding the underlying mechanisms governing protein fouling during virus filtration and there is no quantitative understanding of the effects of solution pH on the overall performance characteristics of these virus filters. The objective of this study was to obtain quantitative data for the effect of solution pH on protein transmission, filtrate flux, and membrane capacity during BSA filtration through the Viresolve 180 membrane and to use these results to develop a more fundamental understanding of the fouling/filtration mechanisms.

Materials and Methods

Solution Preparation

Solutions of 0.01 M KCl were prepared by dissolving preweighed amounts of KCl (J.T. Baker, Philipsburg, NJ) in deionized distilled water obtained from a NANOpure® water purification system (Model D11911, Barnstead/Thermodyne Co., Dubuque, IA) with resistivity greater than 18 MΩ-cm. Salt solutions were buffered with 0.001 M phosphoric acid (J.T. Baker) for pH between 3 and 5.5, and with potassium phosphate monobasic (Sigma, St. Louis, MO) for pH 5.8–8. All buffer solutions were prefiltered through 0.2 μm pore-size Supor®-200 membranes (PALL Corp., Ann Arbor, MI) to remove any particles or undissolved salts prior to use.

BSA (Sigma A7906, Lot No. 066K0708) was used as a model protein due to the extensive previous literature of BSA fouling during ultrafiltration and microfiltration. The Sigma BSA contains about 15% dimers (Kelly et al., 1993) and a small concentration (approximately 0.03% by mass) of large aggregates (Ho and Zydney, 1999). Protein solutions were prepared by dissolving the appropriate mass of powdered protein in buffered 0.01 M KCl. The pH was adjusted using small amounts of 1 M HCl or KOH as required and was measured with a model 420Aplus Thermo Orion pH meter (Orion Technology, Beverly, MA). Protein solutions were prepared fresh for each filtration experiment and were always used within 6 h of preparation. All protein solutions were filtered through a 0.22 μm Acrodisc® syringe filter (PALL Corp.) to remove any large insoluble protein aggregates immediately prior to use. Protein concentrations were analyzed using a UV-vis spectrophotometer (SPECTRAmax plus384, MD Corp., Sunnyvale, CA) with the absorbance measured at 280 nm. Calibration curves were established at each pH using protein solutions of known concentration.

Membranes

All filtration experiments were performed using Viresolve 180 membranes (Millipore Corp., Billerica, MA). The Viresolve 180 membrane has a composite structure made from hydrophilized polyvinylidene fluoride (PVDF). The membrane has a thin skin layer that provides the desired virus retention and is designed for permeation of proteins with molecular weight less than 180 kDa. Each membrane disk was soaked for 24 h at 4°C in a 3 g/L protein solution in 0.01 M KCl at pH 4.7 (the isoelectric point of BSA) to insure equilibrium adsorption prior to use. Immediately before the experiment, each membrane was gently rinsed with 0.01 M

KCl at pH 7.0 to remove any labile protein. A single membrane was used for each filtration experiment.

Protein Filtration

All filtration experiments were conducted in a 10 mL Amicon stirred ultrafiltration cell (Model 8010, Millipore Corp., Bedford, MA). Membranes were cut into 25 mm disks from a large flat sheet using a specially designed cutting device. A 25 mm diameter Tyvek spacer (pore size of approximately 30 μm) was placed beneath the membrane to prevent deformation of the membrane structure into the support. Data were obtained with the membrane oriented so that the skin-side was either up (facing the feed) or down (substructure facing the feed) to obtain additional insights into the mechanisms governing fouling and protein transmission.

Filtration experiments were performed at either constant pressure, which was maintained by air pressurization of the stirred cell, or at constant filtrate flux. Constant flux was provided by a Masterflex 7523-20 peristaltic pump (Cole-Parmer, Vernon Hills, IL) connected to the filtrate line (downstream of the membrane) to avoid the possibility of protein denaturation or aggregation associated with the pump. The feed reservoir was air-pressurized so that a positive gauge pressure was maintained throughout the system. Filtrate flow rate was measured by timed collection using a digital balance (Model AB104-S, Mettler Toledo, Columbus, OH). Filtrate samples (approximately 200 μL) were collected periodically for measurement of the protein concentration.

The stirred cell and feed reservoir were initially filled with 0.01 M KCl at pH 7.0 taking care to remove any trapped air bubbles from the cell and associated tubing. Data were obtained both with and without stirring, with the stirrer speed adjusted using a Strobotac Type 1531 strobe light (General Radio Co., Concord, MA). A minimum of 40 mL of buffer was flushed through the membranes to wet the pore structure. The clean membrane hydraulic permeability:

$$L_p = \frac{\mu J_v}{\Delta P} \tag{1}$$

was determined from the slope of data for the filtrate flux (J_v) as a function of applied pressure (ΔP) between 7 and 103 kPa (corresponding to between 1 and 15 psi). The stirred cell and feed reservoir were then emptied and refilled with a 1 g/L protein solution at the desired pH. The device was then re-pressurized to begin the protein filtration. After completion of the filtration experiment, the filtrate port was clamped and a 200 μL sample of the bulk solution was taken directly from the stirred cell. The stirred cell was then carefully emptied, rinsed with 0.01 M KCl, and the membrane hydraulic permeability was reevaluated. All experiments were performed at room temperature (22 ± 3°C).

Membrane Characterization

The sieving characteristics of each membrane were determined from dextran sieving measurements obtained in the stirred cell at a stirring speed of 600 rpm. A 4 g/L polydisperse dextran with average molecular weight of 75 kDa (Sigma D1390) was dissolved in a 0.15 M KCl solution buffered with 1 mM potassium phosphate monobasic (Sigma) at pH 7. Filtration was performed at a constant flux of 5 μm/s, maintained with a peristaltic pump on the filtrate line. Dextran solutions were prefiltered through a 0.2 μm pore size membrane to insure that there were no particles or undissolved material in the solution. Filtrate samples were collected after filtration of a minimum of 500 μL to wash out the dead volume beneath the stirred cell. The molecular weight distributions for the dextrans in the bulk and permeate solutions were determined by size exclusion chromatography (Agilent 1100 series, Agilent Technologies, Palo Alto, CA) using a TSK-GEL G4000SW gel filtration column (Tosoh Biosciences, Montgomeryville, PA). The column was calibrated with narrow molecular mass dextran standards ranging from 11 to 165 kDa (American Polymer Standards, Mentor, OH) dissolved in the KCl buffer. Additional details on the dextran analysis are provided by Burns and Zydney (2001).

Results and Discussion

Skin-Side Up Orientation

Typical experimental data for the filtrate flux during the constant pressure ($\Delta P = 103$ kPa = 15 psi) filtration of a 1 g/L BSA solution through a Viresolve 180 membrane with the skin-side up are shown in the top panel of Figure 1 for several values of the solution pH. The error bars on the measured values were all within the size of the symbols and are thus not shown for clarity. Repeat measurements under the same conditions were reproducible, with deviations in the flux of less than 10%. The net protein charge, as determined by Menon and Zydney (1998) using capillary electrophoresis, varied from +11 at pH 4 to −18 at pH 7. The initial flux was lowest at pH 4.7 ($J_v = 21$ μm/s = 76 L/m^2/h), which is the isoelectric point of BSA, and increased to more than 100 μm/s (360 L/m^2/h) at pH 7. The flux declined quite rapidly at the start of the filtration, decreasing to 20% of its initial value after 20 min at pH 7 and 90 min at pH 4.7. This flux decline was directly related to the convective flow during filtration—the hydraulic permeability of the membrane evaluated after protein adsorption was only about 10% less than the value for the clean membrane. In contrast, the permeability measured after the protein filtration at pH 7 was only 67% of the initial value and this dropped to 60% for the experiment performed at pH 4.7.

The corresponding data for the protein concentration in the filtrate stream, C_f, are shown in the bottom panel of Figure 1. The concentrations were measured by collecting

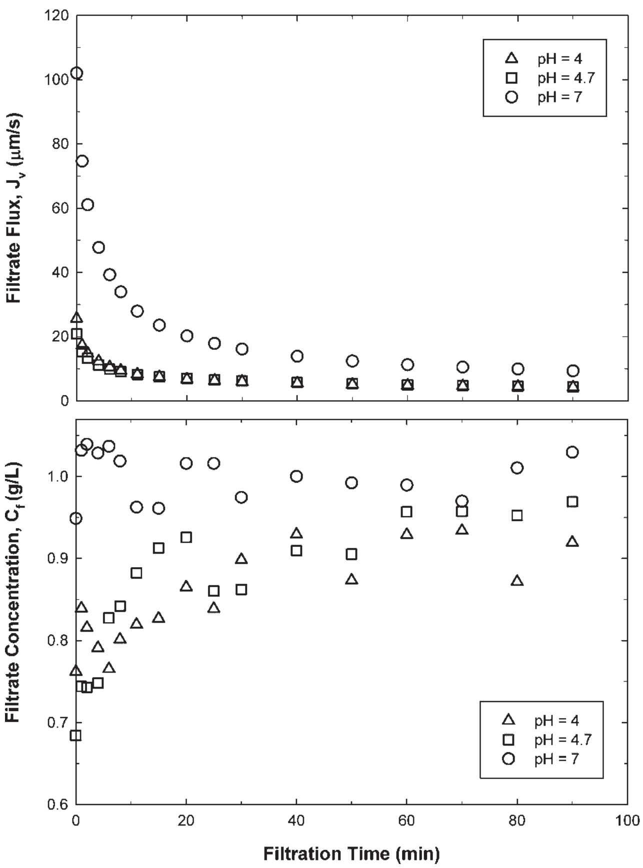

Figure 1. Filtrate flux (**top panel**) and filtrate concentration (**bottom panel**) as a function of time for the unstirred filtration of a 1 g/L solution of BSA at different pH through the Viresolve 180 membrane with the skin-side up at a constant pressure of 103 kPa.

small samples (approximately 200 μL) at each time point. The filtrate concentration at pH 7 was fairly constant throughout the filtration with a value approximately equal to the feed concentration $C_F = 1$ g/L. In contrast, the filtrate concentration for the experiments at pH 4 and 4.7 started at a value around 0.7 g/L and gradually increased towards a value of 1 g/L over the course of the 90-min filtration. However, the observed sieving coefficient, defined as the ratio of the filtrate to bulk protein concentrations, remained below 0.95; the measured bulk concentration in the stirred cell at the end of the experiment at pH 4.7 had increased by about 4% due to retention of the protein throughout the course of the filtration.

The performance characteristics of the virus filtration system at different pH were compared more quantitatively by evaluating the membrane capacity and the overall protein yield. The capacity was defined as the total cumulative filtrate volume collected up until the point where the flux had declined to 10% of the clean membrane flux (which was $J_o = 136 \pm 10$ μm/s for these experiments). The calculated values of the capacity from the data in Figure 1 range from only 3 L/m^2 for the filtration at the protein isoelectric point, compared to a value of 75 L/m^2 at pH 7 (see Table I). The effect of solution pH on the capacity is shown more explicitly in Figure 2. The membrane capacity showed a weak minimum at the protein isoelectric point and then increased quite sharply up until pH 7.

The corresponding values of the protein yield (Y) for these experiments were evaluated as:

$$Y = \frac{\int_0^t C_f J_v \, dt}{C_F \int_0^t J_v \, dt} \tag{2}$$

where C_F is the protein concentration in the feed (equal to the initial bulk protein concentration). Equation (2) was integrated numerically using Simpson's rule up until the time at which $J_v/J_o = 0.1$, with the results summarized in Table I. The protein yield was only 75% at pH 4.7 but was nearly 100% at pH 7.

In order to understand the flux and sieving behavior in more detail, the membranes used for the experiments in Figure 1 were gently rinsed and then used to evaluate the dextran sieving coefficients. Data were obtained using a 4 g/L dextran solution at a constant flux of 5 μm/s, with this relatively low filtrate flux used to minimize the effects of concentration polarization. Results are shown in Figure 3 along with data obtained for a clean membrane and for a membrane that had been presoaked in a 3 g/L solution of BSA at pH 4.7 for 24 h at 4°C. The dextran sieving coefficients for the membrane that had been preadsorbed with BSA are only slightly smaller than those for the clean membrane, consistent with the relatively small reduction in permeability seen after protein adsorption. There was a further reduction in the dextran sieving coefficients for the fouled membranes. For example, the sieving coefficient of a 50 kD dextran was $S_o = 0.43$ for the clean membrane compared to values of 0.15 for the membrane used to filter the BSA at pH 4.7 and 0.20 for the membrane used at pH 7. The difference in the dextran sieving coefficients for the membranes used at pH 4.7 and 7 was relatively small (typically less than 25%), particularly compared to the very

Table I. Membrane capacity and protein yield at different pH for filtration of 1 g/L BSA solutions through the Viresolve 180 membrane with the skin-side up and skin-side down orientations.

	Membrane capacity (L/m^2)[a]		Protein yield (%)	
pH	Skin-side up	Skin-side down	Skin-side up	Skin-side down
4	5	110	80	100
4.7	3	27	75	100
7	75	240	100	100

[a]The capacity was defined as the total cumulative filtrate volume collected before the flux decreased to 10% and 30% of the value for the clean membrane in skin-side up and skin-side down orientations, respectively.

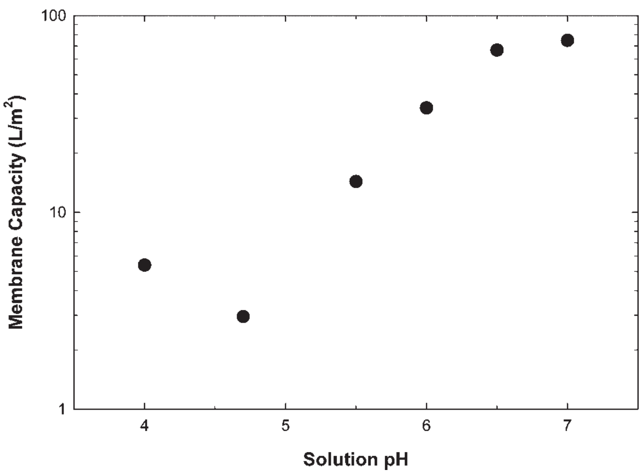

Figure 2. Membrane capacity as a function of solution pH for the unstirred filtration of a 1 g/L solution of BSA through the Viresolve 180 membrane with the skin-side up at a constant pressure of 103 kPa.

large difference in filtrate flux and membrane capacity seen in Figure 1. These results are discussed in more detail subsequently.

Concentration Polarization Effects

The relatively small reduction in the membrane permeability and dextran sieving coefficients after protein filtration suggests that the flux decline observed in Figure 1 is due not to some type of irreversible fouling but might instead be associated with concentration polarization effects during

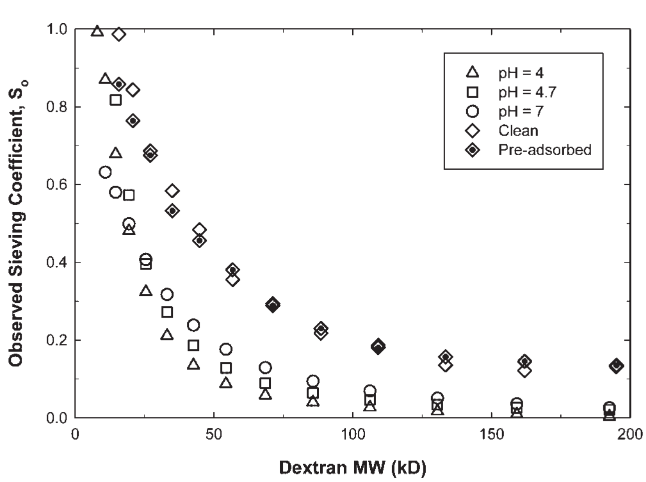

Figure 3. Dextran sieving coefficients for clean, preadsorbed, and fouled membranes (at different pH) at constant flux of 5 μm/s.

normal flow filtration. This phenomenon was examined by performing a series of experiments in the presence of stirring (stirring speed of 600 rpm) to increase the rate of back mass transfer and thus control the extent of polarization in the stirred cell. Results for the filtrate flux and filtrate concentration are shown in the top and bottom panels of Figure 4. The initial values of the flux were very similar to those obtained in the unstirred experiments, particularly at pH 4 and 4.7. However, the rate of flux decline was much less pronounced. The flux at pH 4.7 remained above 13 μm/s (47 L/m^2/h) throughout the 40-min filtration compared to the 70% reduction in flux seen previously in Figure 1. The net result was that the membrane capacity at the protein isoelectric point with the skin-side up was 28 L/m^2 in the presence of stirring compared to only 3 L/m^2 in the same orientation but without stirring. Similarly, the flux at pH 7 decreased by about 50% after 40 min of filtration compared to the 90% reduction in flux obtained in the unstirred experiment. The filtrate concentration (bottom panel) decreased with time during the stirred cell filtration, which is exactly opposite of the behavior seen in Figure 1 for the unstirred system. The filtrate concentration at pH 7 decreased to about 0.3 g/L at $t = 40$ min filtration, while the filtrate concentration at pH 4.7 decreased to a value of less than 0.1 over most of the experiment.

The experimental data in Figure 4 were re-plotted in Figure 5 in the form suggested by the simple stagnant film model for concentration polarization (Zeman and Zydney, 1996):

$$J_v = k_m \ln\left[\frac{C_w - C_f}{C_b - C_f}\right] \tag{3}$$

where k_m is the bulk mass transfer coefficient and C_w is the protein concentration in the solution immediately adjacent to the membrane surface. The bulk protein concentration, C_b, was evaluated as a function of time by numerical integration of the protein mass balance over the stirred cell (using the experimental values of J_v and C_f):

$$V\frac{dC_b}{dt} = J_v A(C_F - C_f) \tag{4}$$

where V is the constant volume of the stirred cell and A is the effective filtration area. The semi-log plots are highly linear with r^2 values greater than 0.95 at all pH, consistent with predictions of the concentration polarization model (Eq. 3) with a constant value of $C_w - C_f$. The assumption of constant C_w is consistent with the high pressure used for the filtration; this should cause the protein osmotic pressure difference across the membrane to remain nearly constant at a value slightly below ΔP. The filtrate concentration C_f is always small relatively to C_w and thus has no significant affect on the analysis.

The slopes for the semi-log plots in Figure 5 should be equal to the protein mass transfer coefficients, with the values of k_m (determined by simple linear regression of the

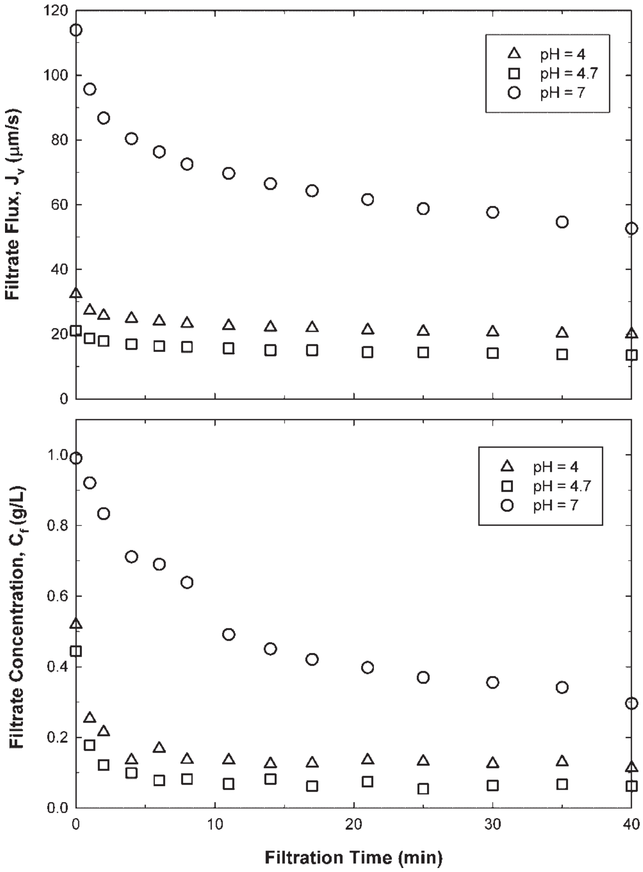

Figure 4. Filtrate flux (**top panel**) and filtrate concentration (**bottom panel**) as a function of time for the stirred cell filtration of a 1 g/L solution of BSA at different pH through the Viresolve 180 membrane with the skin-side up at a constant pressure of 103 kPa.

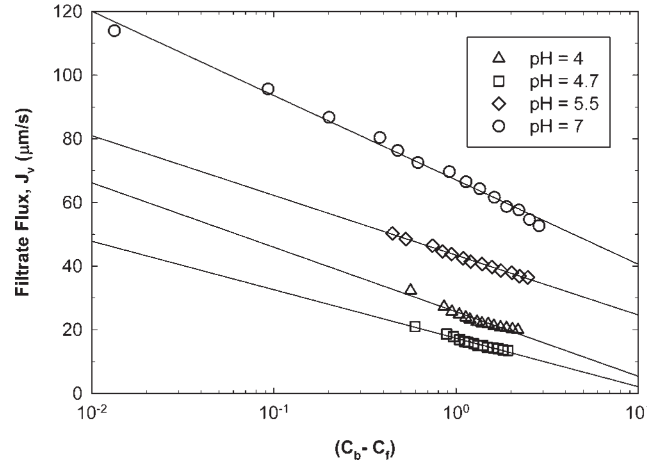

Figure 5. Filtrate flux as a function of the concentration driving force at different pH.

data) shown in the fourth column of Table II. The mass transfer coefficient is minimum at the protein isoelectric point (pH 4.7) with a value of 6.6×10^{-6} m/s and increases by more than 70% as the pH increased to 7. This strong dependence of the mass transfer coefficient on pH is consistent with available data for the pH dependence of the protein diffusion coefficient (Doherty and Benedek, 1974) which show a twofold increase in the diffusivity as the pH goes from 4.7 to 7. In addition, the results are in good agreement with available correlations for the mass transfer coefficient in a stirred cell (Smith et al., 1968):

$$k_\mathrm{m} = \chi\left(\frac{\omega b^2}{\upsilon}\right)^{0.567}\left(\frac{\upsilon}{D}\right)^{0.33}\frac{D}{b} \tag{5}$$

where ω is the stirring rate, ν is the kinematic viscosity, D is the protein diffusion coefficient, and b is the radius of the stirred cell. The coefficient χ is a function of the detailed device geometry and impeller location. Opong and Zydney (1991) evaluated χ as 0.23 based on filtrate flux data through fully retentive ultrafiltration membranes. The calculated values of the mass transfer coefficients, determined using literature data for the BSA diffusion coefficient (Doherty and Benedek, 1974), are in good qualitative agreement with the experimental results, providing further confirmation of the importance of concentration polarization effects during protein filtration through the Viresolve 180 membranes. The calculated mass transfer coefficients are uniformly smaller than the experimental values, which may simply be due to differences in impeller location (or geometry) in combination with differences in membrane porosity, both of which would affect the coefficient χ.

The final column of Table II shows the calculated values of the wall concentration determined from the intercepts of the data in Figure 5. The wall concentration was minimum at the protein isoeletric point (pH 4.7) with a value of about 15 g/L and increased by more than a factor of 20 as the pH increased to 7. These large values of the wall concentration imply that the actual protein sieving coefficients through the Viresolve 180 membrane are quite small. For example, the filtrate concentration data in Figure 4 at $t = 40$ min correspond to actual sieving coefficients:

$$S_\mathrm{a} = \frac{C_\mathrm{f}}{C_\mathrm{w}} \tag{6}$$

of 0.003, 0.005, and 0.0009 at pH 4, 4.7, and 7, respectively. The actual sieving coefficient is maximum around the protein isoelectric point, which is consistent with previous studies of BSA transmission through ultrafiltration membranes and is a direct result of the lack of significant electrostatic exclusion at pH 4.7 due to the absence of any net electrical charge on the BSA under these conditions.

Table II. Mass transfer parameters for filtration of BSA.

pH	Diffusion coefficient ($\times 10^{11}$ m^2/s; Doherty and Benedek, 1974)	Calculated mass transfer coefficient ($\times 10^6$ m/s) [Eq. 5]	Experimental mass transfer coefficient (m/s $\times 10^6$) [Fig. 5]	Wall Concentration, C_w (g/L)
4	—	—	8.8	20
4.7	7.7	6.0	6.6	15
5.5	9.6	7.0	8.2	200
7	16	9.8	11.5	340

Skin-Side Down Orientation

Most current virus filtration applications use membranes oriented with the selective (skin) layer faced away from the feed stream. Experimental data for filtrate flux and filtrate concentration for the filtration of a 1 g/L BSA solution through the Viresolve 180 membrane with the skin-side down are shown in the top and bottom panels of Figure 6. The filtrate flux with the skin-side down orientation are uniformly higher than the values obtained with the skin-side up for normal flow filtration (Fig. 1) and are actually similar to the results obtained previously with the skin-side up and in the presence of stirring (Fig. 4). For example, at $t = 40$ min, the flux at pH 7 with the skin-side down is 54 μm/s compared to values of 14 μm/s for the skin-side up orientation in the absence of stirring and 53 μm/s for the skin-side up but with stirring. In addition, the hydraulic permeability measured after the protein filtration at pH 7 was 69% of the initial value and this increased to 87% for the experiment performed at pH 4.7. These values were both slightly higher than the corresponding values obtained in the skin-side up orientation, indicating that there was less irreversible fouling when the flow enters the membrane through the substructure.

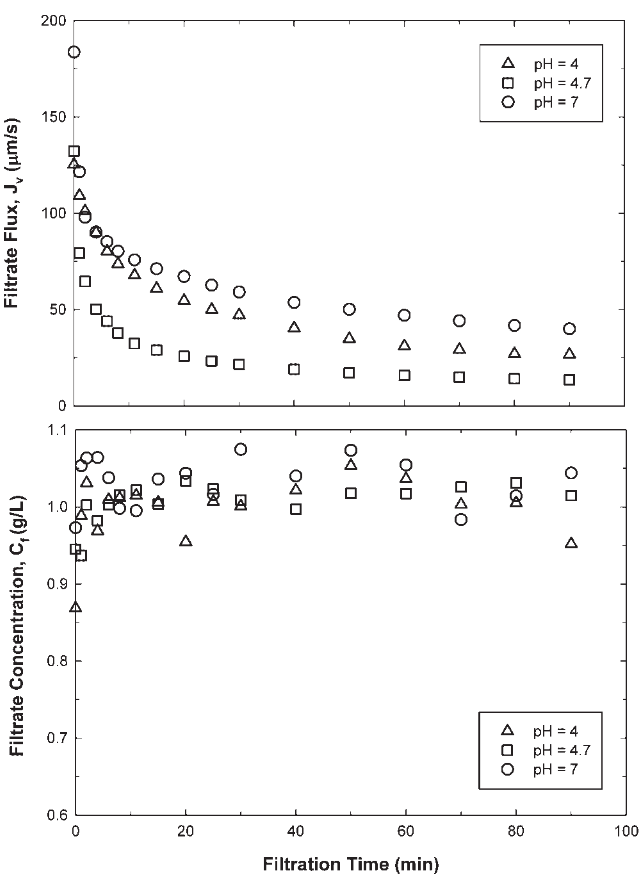

Figure 6. Filtrate flux (**top panel**) and filtrate concentration (**bottom panel**) as a function of time for the unstirred filtration of a 1 g/L solution of BSA at different pH through the Viresolve 180 membrane with the skin-side down at a constant pressure of 103 kPa.

The filtrate concentration data for the skin-side down experiments remained fairly constant throughout the filtration with a value approximately equal to the feed concentration at all three solution pH. The large values of the observed sieving coefficient ($S_o = C_f/C_b$) are consistent with the high degree of internal concentration polarization within the membrane substructure that occurs when the membrane is operated in the skin-side down orientation. This type of internal polarization has been discussed previously by Syedain et al. (2006).

The calculated values of the membrane capacity and protein yield for the experiments in Figure 6 are summarized in Table I. In this case, the capacity was defined as the total cumulative filtrate volume collected before the flux decreased to 30% of the value for the clean membrane since the flux had not dropped much below this level even after 90 min of filtration. The protein yield was evaluated from Equation (2) with the integration performed over the same time interval used for the capacity. The membrane capacity with the skin-side down orientation was much larger than that with the skin-side up; the capacity at pH 4 increased by more than 20-fold simply by switching the membrane orientation and this effect would have been even more pronounced had the filtration (and thus the capacity) with the skin-side down been evaluated out until the flux had decayed to 10% of the value for the clean membrane. The capacity was again a strong function of solution pH, with the capacity at pH 7 (240 L/m^2) being more than eight times the value at the protein isoelectric point. The protein yields in the skin-side down orientation, which were calculated by integrating Equation (2) up until the time at which $J_v/J_o = 0.3$, are essentially 100% irrespective of the solution pH. The high yield in the skin-side down orientation is due to the high degree of protein transmission arising from the extensive internal concentration polarization that occurs within the substructure when the membrane is oriented with the skin-side facing away from the feed.

The membranes used for the experiments in Figure 6 were removed from the stirred cell, gently rinsed, and then returned to the stirred cell but now oriented with the skin-side up. The stirred cell was then filled with a dextran solution, with the dextran sieving coefficients evaluated for the fouled membranes at a stirring speed of 600 rpm. This eliminated any affects associated with concentration polarization within the membrane substructure. The sieving curves were qualitatively similar to those seen in Figure 3 for the membranes used with the skin-side up. Figure 7 shows data for the sieving coefficient of a single dextran (MW = 50 kDa), with each bar representing data for a separate Viresolve 180 membrane used for a particular protein filtration (skin-up unstirred, skin-up stirred, or skin-down) and at a specific pH (4, 4.7, or 7). The sieving coefficients for the membranes used in the skin-side down orientation were considerably larger than those obtained for the membrane used with the skin-side up (in the absence of stirring), particularly at pH 4 and 4.7. For example, the sieving coefficient of the 50 kDa dextran at pH 4.7 through the membrane oriented with the skin-side down was $S_o = 0.29$ compared to a value of 0.15 with the skin-side up. The greatest dextran sieving coefficients were seen with the membranes that had been used with the skin-side up in the presence of stirring, although this difference was probably in large part to the much shorter filtration time used for the stirred experiments ($t = 40$ min) compared to the 90-min filtration for the experiments in which the skin-side was oriented away from the feed (or with the skin up but in the absence of stirring).

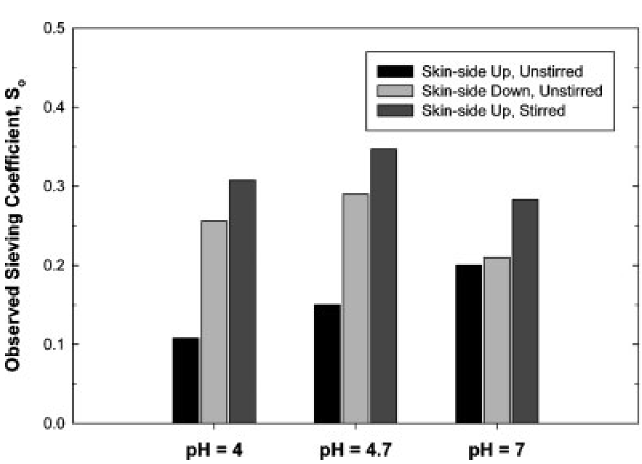

Figure 7. The observed sieving coefficient of a 50 kDa dextran through membranes fouled at different pH and under different filtration conditions (unstirred skin up, stirred skin up, and unstirred skin down).

Conclusions

The experimental data obtained in this study provide the first quantitative results for the effects of solution pH on the performance characteristics of virus filtration systems. The data clearly demonstrate that membrane capacity is minimum near the protein isoelectric point and increases monotonically with pH at pH above the isoelectric point. The low flux obtained at pH 4.7 was due primarily to the small value of the protein mass transfer coefficient under these conditions, an effect that is directly related to the small value of the protein diffusion coefficient near the isoelectric point. The reduction in protein mass transfer coefficient appeared to be more important than the small increase in membrane resistance due to the slightly greater amount of protein fouling at pH 4.7. The protein yield was close to 100% when the membrane was used in the skin-side down orientation due to the high degree of concentration polarization within the membrane support structure. In contrast, the protein yield with the skin-side up was only 80% at the protein isoelectric point but increased to essentially 100% at pH 7.

The filtrate flux data obtained in the skin-side up orientation were very consistent with predictions of the concentration polarization model, both in terms of the linear dependence on the concentration driving force and the calculated values of the mass transfer coefficient. However, it is important to note that the estimated wall concentration at pH 4.7 ($C_w = 15$ g/L) is much too low to provide the osmotic pressure needed to explain the relatively low flux obtained under these conditions. This apparent discrepancy could be due to the formation of a reversible "gel layer" on the membrane surface which would provide a significant additional hydraulic resistance to filtration. There was no evidence of a gel layer in either the measured values of the membrane permeability or dextran sieving coefficients after protein filtration, although it is possible that this layer could have been re-solubilized during the rinsing of the stirred cell used to remove the bulk protein solution or during the post-sieving permeability experiments. Future studies will be needed to examine this behavior in more detail and to develop appropriate models for the flux during virus filtration.

The effect of pH on the membrane capacity could have significant implications for the design of downstream processes for protein purification. Virus filtration is typically done after the primary purification, for example, after a Protein A affinity column in the purification of monoclonal antibodies. But, the exact location of the virus filtration in the overall process is somewhat flexible relative to the other chromatography and purification steps. The data obtained in this study clearly demonstrate that the capacity of the virus filter can be significantly improved by performing the filtration at pH well removed from the protein isoelectric point, conditions that may well occur naturally at specific points within the downstream process due to the different elution conditions used in the various chromatography steps. The possibility of using the solution pH, based on the location in the overall downstream process, for optimization of the virus filtration has not been discussed previously in the open literature.

Although the protein concentration in the filtrate solution was greatest during filtration at pH 7, the calculated value of the actual protein sieving coefficient was minimum under these conditions. The higher degree of protein retention at pH 7 is consistent with the increase in electrostatic exclusion associated with the greater electrical charge on BSA at pH well above the protein isoelectric point, a phenomenon that has been discussed in some detail by Pujar and Zydney (1994) and Burns and Zydney (1999). The small values of the actual sieving coefficient at pH 7 did not cause a loss of protein yield because the large filtrate flux caused a very high degree of concentration polarization under these conditions.

The membrane capacity and protein yield were both significantly higher when the membrane was oriented with the skin-side down, even though the capacity was defined as the total cumulative filtrate volume collected before the flux decreased to 30% of the value for the clean membrane (in contrast to 10% when the membrane was oriented with the skin-side up). The capacity difference was most pronounced at pH 4 where the change in orientation caused more than a 20-fold increase in capacity. The high protein yield in the skin-side down orientation is consistent with the high degree of protein polarization that occurs within the membrane support structure under these conditions. However, the high filtrate flux (and capacity) in the skin-side down orientation is somewhat unexpected; the high degree of internal concentration polarization should reduce the flux due to the high protein osmotic pressure. Instead, the flux profile in the skin-side down orientation looked very similar to that seen with the skin-side up but in the presence of stirring, which significantly reduces the extent of concentration polarization. The dextran sieving data clearly showed that there was less fouling in the skin-side down orientation, suggesting that the substructure may somehow protect the skin layer from fouling when operated in this orientation. Future studies will be required to clarify the origin of this phenomenon and to quantify its effect on filter performance during virus filtration.

Nomenclature

A	effective filtration area (m^2)
b	radius of the stirred cell (m)
C_b	protein concentration in the bulk (g/L)
C_F	protein concentration in the feed (g/L)
C_f	protein concentration in the filtrate (g/L)
C_w	protein concentration at the membrane surface (g/L)
D	protein diffusion coefficient (m^2/s)
J_o	buffer flux through clean membrane (m/s)
J_v	filtrate flux (m/s)
k_m	bulk mass transfer coefficient (m/s)
L_p	membrane hydraulic permeability (m)
S_a	actual sieving coefficient
t	filtration time (s)
V	volume of the stirred cell (m^3)
Y	protein yield
ΔP	transmembrane pressure (Pa)
χ	mass transfer coefficient parameter
μ	viscosity (kg/m/s)
ν	kinematic viscosity (m^2/s)
ω	stirring rate (1/s)

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